

Project Specification: "SmartRetail 360" Analytics & Support Platform

Project Overview

You are tasked with building **SmartRetail 360**, an end-to-end data and AI intelligence platform for a scaling e-commerce business. The platform will process raw business data, forecast sales using machine learning, and leverage Generative AI to allow customer support agents to ask conversational questions about product reviews.

You have **2-3 weeks** to complete this capstone. The project must be fully functional, modular, and hosted on GitHub.

Phase 1: Data Engineering & Predictive Analytics

Milestone 1: The Data Pipeline (ETL)

Your first objective is to build an automated ingestion engine to clean and centralize scattered business data.

- **Ingestion:** Extract raw transaction logs from `sales_data.csv` and unstructured customer reviews from a `reviews.json` file.
- **Data Cleaning:** Use Pandas to handle missing values, drop invalid records, format data types (e.g., convert timestamps), and remove statistical outliers from the sales data.
- **Database Storage:** * Load the structured, cleaned sales data into a relational database (SQLite).
 - Load the semi-structured customer reviews into a document database (MongoDB).

Milestone 2: Exploratory Data Analysis & Machine Learning Modeling

Turn your raw data into active business insights.

- **Visual Analytics:** Generate at least three distinct visualizations (e.g., top-performing products, monthly sales trends, distribution of review ratings) using Matplotlib or Seaborn.
- **Feature Engineering:** Prepare your data for modeling by scaling numerical variables and encoding categorical columns. Split your data into distinct training and testing sets.
- **Predictive Modeling:** Train a **Linear Regression** model designed to predict sales revenue for the upcoming month.
- **Model Evaluation:** Calculate and output key evaluation metrics (such as Mean Squared Error or R^2 score) to verify the accuracy of your model.

Phase 2: Generative AI, APIs & Application Interface

Milestone 3: Vector Embeddings & Semantic Search

Create a searchable knowledge base out of the customer feedback records stored in MongoDB.

- **Embedding Generation:** Pull product reviews from your database and process them through an embedding model (via Hugging Face or OpenAI) to convert text reviews into dense vector representations.
- **Knowledge Storage:** Store these vector embeddings into a local indexing system to facilitate rapid semantic vector searches.

Milestone 4: Backend API Development & RAG Orchestration

Build the intelligence layer that coordinates your models and serves them over the network.

- **RAG Implementation:** Utilize **LangChain** to connect your vector store with a Large Language Model. Implement **Prompt Engineering** to ensure the LLM references the retrieved review context accurately and maintains a professional tone.
- **API Deployment:** Build a **FastAPI** backend application that exposes two operational endpoints:
 - **POST /predict-sales:** Accepts features and returns the ML model's revenue forecast.
 - **POST /ask-assistant:** Accepts a natural language question, queries the vector store for context, and returns the LLM's summarized answer.

Milestone 5: Web Application Interface & Final Delivery

Tie the entire system together into a clean, intuitive dashboard.

- **Streamlit UI:** Construct a frontend interface featuring two core navigation views:
 1. *Business Dashboard View:* Displays your historical data charts alongside an interactive form to trigger and view the machine learning sales predictions.
 2. *AI Assistant View:* An interactive chat interface that communicates with your **/ask-assistant** endpoint, enabling users to query review trends conversationally.
- **Testing & Deployment:** Run integration tests to ensure data flows flawlessly from the databases through the APIs to the frontend UI.
- **Portfolio Submission:** Structure your code repository cleanly, draft an informative **README.md** containing an architecture overview and setup steps, and push your final code to **GitHub**.

Final Project Deliverables

1. **Source Code:** A fully functional repository hosting your ETL scripts, ML training code, FastAPI backend, and Streamlit frontend.
2. **Database Artifacts:** Seed scripts or local setup steps for your SQLite database and MongoDB collections.
3. **Documentation (README.md):** Clear descriptions of how to install dependencies, run the applications, and interact with the endpoints.